Partial Fractions and Improper Integrals

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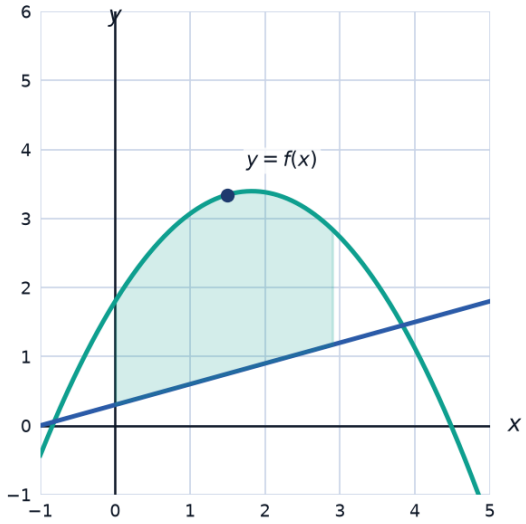
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Partial fraction decomposition: distinct linear factors

Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions.

2 [Foundational] • Target: Partial fraction decomposition: distinct linear factors

Evaluate $\int \frac{3x+5}{x^2+x-2}\,dx$ using partial fractions (note $x^2+x-2=(x+2)(x-1)$).



3 [Foundational] • Target: Partial fraction decomposition: distinct linear factors

Decompose $\frac{x+7}{x^2-x-6}$ into partial fractions (note $x^2-x-6=(x-3)(x+2)$).

4 [Foundational] • Target: Partial fractions with a repeated linear factor

Evaluate $\int \frac{2x+3}{(x-1)^2}\,dx$ using the decomposition found above.

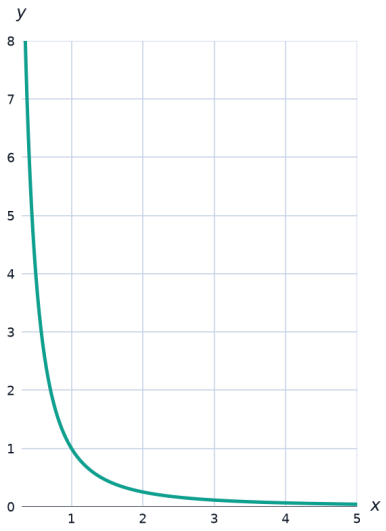
5 [Foundational] • Target: Partial fractions with an irreducible quadratic factor

Set up (but do not solve for constants) the partial fraction decomposition of $\frac{3x^2+2}{(x-1)(x^2+4)}$.

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6 [Foundational] • Target: Improper integrals: infinite limits

Evaluate $\int_1^{\infty} \frac{1}{x^2}\,dx$, or show it diverges.



7 [Foundational] • Target: Improper integrals: discontinuous integrands

Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$, or show it diverges (note the integrand is undefined at $x = 0$).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Improper integrals: discontinuous integrands

Evaluate $\int_0^2 \frac{1}{(x-2)^2}\,dx$, or show it diverges (the integrand is undefined at $x = 2$).

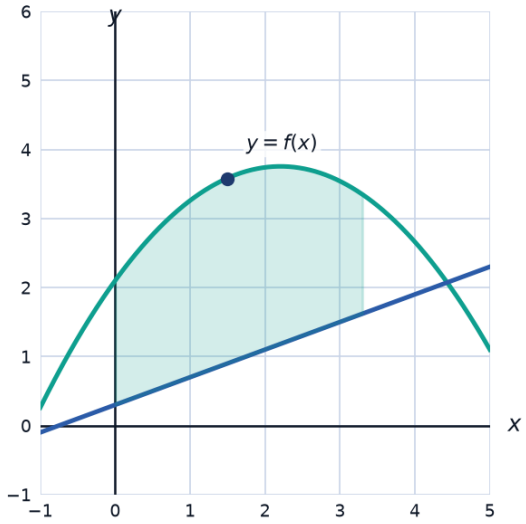
9 [Intermediate] • Target: The p-test for improper integrals

State the p-test for $\int_1^{\infty} \frac{1}{x^p}\,dx$, and use it to determine convergence for $p = 3$ and $p = 0.5$.

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10 [Intermediate] • Target: The p-test for improper integrals

State the analogous p-test for $\int_0^1 \frac{1}{x^p}\,dx$ (a discontinuity at $x = 0$), and determine convergence for $p = \frac{1}{2}$ and $p = 2$.



11 [Intermediate] • Target: Combining partial fractions with improper integrals

Evaluate $\int_0^\infty \frac{1}{(x+1)(x+2)}\,dx$ using partial fractions.

12 [Intermediate] • Target: Three distinct linear factors

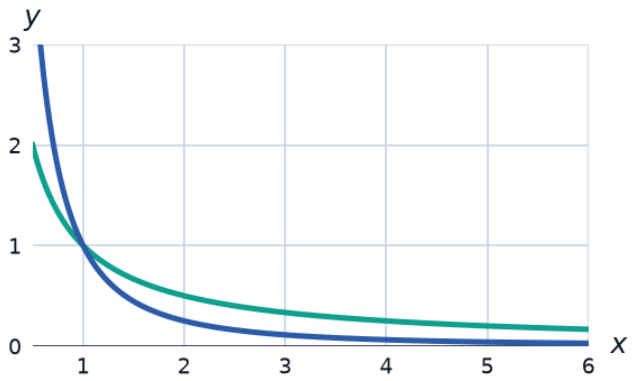
Evaluate $\int \frac{6x^2 - 3x - 1}{x(x-1)(x+1)}\,dx$ using the decomposition above.

13 [Intermediate] • Target: Visualizing convergence: area under the curve

The convergent improper integral $\int_1^\infty \frac{1}{x^2}\,dx = 1$ (shown earlier) means the infinite region under $y = \frac{1}{x^2}$ for $x \geq 1$ has FINITE area. Explain intuitively why this is possible even though the region extends forever.

14 [Intermediate] • Target: Visualizing convergence: area under the curve

By contrast, $\int_1^\infty \frac{1}{x}\,dx$ diverges. Compare the graphs of $y = \frac{1}{x}$ and $y = \frac{1}{x^2}$ on $[1, 10]$ and explain why $\frac{1}{x}$ decays too slowly for the area to stay finite.



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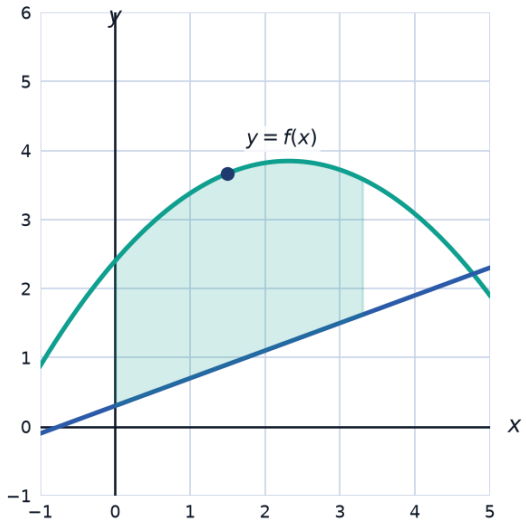
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Improper integrals with both an infinite bound and a discontinuity • Cross-Lesson: 4.02 Area in Polar Coordinates

Evaluate $\int_0^{\infty} \frac{1}{\sqrt{x}(1+x)}\,dx$ is a classic doubly-improper integral (discontinuous at $x = 0$ AND infinite upper bound). Explain how such an integral must be split to evaluate it.

16 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Use a direct comparison to argue that $\int_1^{\infty} \frac{1}{x^2+1}\,dx$ converges, without computing it exactly.



17 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Use a direct comparison to argue that $\int_1^{\infty} \frac{2+\sin x}{x}\,dx$ diverges.

18 [Challenging] • Target: Combining partial fractions with improper integrals • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Level 3 · Advanced / AP FRQThe alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$ converges to sum S. If the sum is approximated by the fourth partial sum S_4, use the Alternating Series Error Bound to find an upper bound for the error $|S - S_4|$.

19 [Challenging] • Target: Three distinct linear factors • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Level 2 · MediumApply the Ratio Test to determine the convergence or divergence of $\sum_{n=1}^{\infty} \frac{4^n}{n!}$.

20 [Challenging] • Target: Visualizing convergence: area under the curve • Cross-Lesson: 1.01 Integration by Parts

Level 1 · FoundationsExpress the repeating decimal $0.272727\ldots = 0.\overline{27}$ as an infinite geometric series, and find its exact fraction representation in simplest form.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Partial fraction decomposition: distinct linear factors

Use a direct comparison to determine whether $\int_1^{\infty} \frac{4+\sin x}{x}\,dx$ converges or diverges.

a. Diverges only because $a = 4$ happens to be even.

b. Diverges, since $\frac{4+\sin x}{x} \geq \frac{3}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{3}{x}\,dx$ diverges since $p = 1 \leq 1$.

c. Converges, since $\frac{4+\sin x}{x} \leq \frac{5}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{5}{x}\,dx$ converges since $p = 1 > 1$.

d. Cannot be determined by comparison, since $\sin x$ oscillates.

22 [Foundational] • Target: Partial fractions with a repeated linear factor

Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ using partial fractions (telescoping).

a. 2

b. diverges

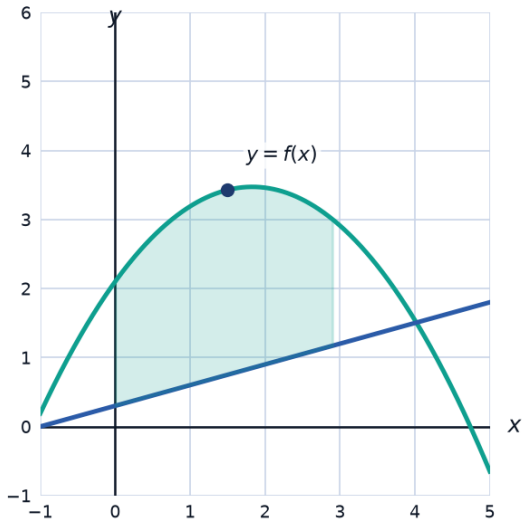
c. 1

d. $\frac{1}{2}$

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23 [Foundational] • Target: Partial fractions with an irreducible quadratic factor

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+6}$ converges.



a. Diverges by the nth – term test

b. Converges (compare to $\sum 1/n^2$)

c. Diverges (compare to $\sum 1/n$)

d. Inconclusive

24 [Foundational] • Target: Improper integrals: infinite limits

Consider the infinite series $\sum_{n=1}^{\infty} \frac{n^p}{5^n}$. The series converges if and only if the parameter p satisfies which of the following conditions? Use the Ratio Test to determine the interval of convergence.

a. $p > 0$

b. $p < 1$

c. $p > -1$

d. For all real values of p

25 [Foundational] • Target: Improper integrals: discontinuous integrands

For $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$, apply the ratio test and determine convergence.

a. Inconclusive ($L = 1$)

b. Converges ($L = 1/2 < 1$)

c. Diverges ($L = 2 > 1$)

d. Diverges by comparison

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26 [Foundational] • Target: The n-test for improper integrals

Apply the ratio test to $\sum_{n=1}^{\infty} \frac{6^n}{n!}$. What is $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$?

a. 0

b. ∞

c. 1

d. 6

27 [Foundational] • Target: Combining partial fractions with improper integrals

Determine whether $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ converges using the ratio test. (variant 1)

a. Converges ($L = 1/e < 1$)

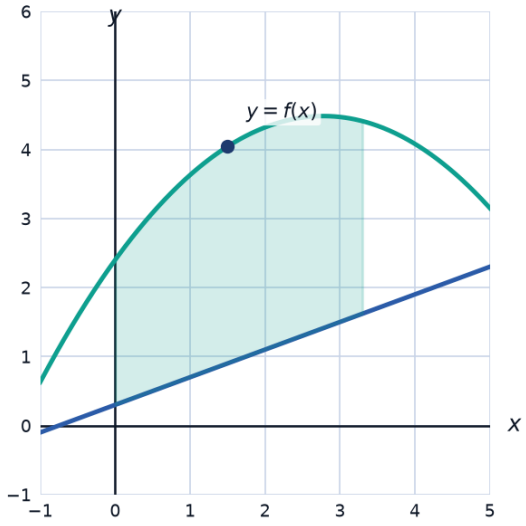
b. Diverges by the nth – term test

c. Diverges ($L = 1/e > 1$)

d. Inconclusive ($L = 1$)

28 [Foundational] • Target: Three distinct linear factors

In the partial fraction decomposition $\frac{1}{(x-3)(x-5)} = \frac{A}{x-3} + \frac{B}{x-5}$, find A.



a. 1

b. $\frac{1}{2}$

c. $\frac{1}{3}$

d. $-\frac{1}{2}$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Visualizing convergence: area under the curve**

Approximating $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ by its first 5 terms, bound the error using the alternating series error bound.

a. $|S - S_5| \leq \frac{1}{49}$

b. $|S - S_5| \leq 36$

c. $|S - S_5| \leq \frac{1}{25}$

d. $|S - S_5| \leq \frac{1}{36}$

30 **[Intermediate]** • **Target: Improper integrals with both an infinite bound and a discontinuity**

Ratio test: if $\lim \frac{a_{n+1}}{a_n} < 1$, the series:

a. Diverges

b. Converges absolutely

c. May converge

d. Converges conditionally

31 **[Intermediate]** • **Target: Comparison test for improper integrals**

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

a. $\frac{1}{x-1} + \frac{1}{(x-1)^2}$

b. $\frac{x}{(x-1)^2}$

c. $\frac{1}{x-1}$

d. $\frac{2}{(x-1)^2}$

32 **[Intermediate]** • **Target: Partial fraction decomposition: distinct linear factors**

Consider the sequence defined by $a_n = \int_0^{1/n} \frac{\sin(x^2)}{x} dx$. Which of the following statements about the convergence of the sequence $\{a_n\}_{n=1}^{\infty}$ is true?

a. The sequence converges to 0.

b. The sequence diverges to infinity.

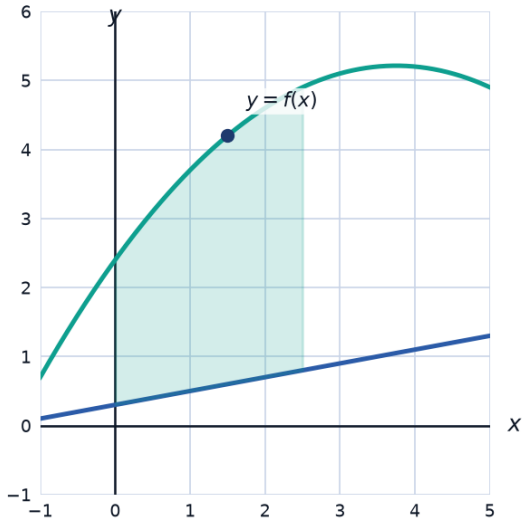
c. The sequence converges to 1.

d. The sequence converges to $\pi/2$.

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33 [Intermediate] • Target: Partial fractions with a repeated linear factor

Consider the infinite series $\sum_{n=1}^{\infty} a_n$. If the terms of the series are given by $a_n = \frac{3n+1}{2n+5}$, which of the following statements about the convergence of the series is true based on the limit of the terms?



a. The series converges because $\lim_{n\to\infty} a_n = 0$.

b. The series diverges because $\lim_{n\to\infty} a_n = \frac{3}{2}$.

c. The series converges because $\lim_{n\to\infty} a_n = \frac{3}{2}$.

d. The series diverges because $\lim_{n\to\infty} a_n$ does not exist.

34 [Intermediate] • Target: Partial fractions with an irreducible quadratic factor

Consider the infinite series $\sum_{n=1}^{\infty} \frac{n^p}{4^n}$. For what values of p does the series converge?

a. All real numbers p

b. $p < 0$

c. $p > 0$

d. No real numbers p

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35 **[Intermediate]** • **Target: Improper integrals: infinite limits**

Consider the sequence $a_n = \frac{1}{n!}\int_0^n e^{-t^2}dt$. Which of the following statements about the convergence of $\{a_n\}$ is true?

a. The sequence diverges to ∞.

b. The sequence converges to 0.

c. The sequence converges to 1.

d. The sequence oscillates and does not converge.

36 **[Intermediate]** • **Target: Improper integrals: discontinuous integrands**

Consider the polar curve given by the equation $r = 3 - 2\sin(\theta)$. A point P on this curve has polar coordinates (r, θ). If the Cartesian coordinates of P are (x, y), and $\theta = \frac{\pi}{2}$, find the value of x.

a. 3

b. 0

c. 1

d. -1

37 **[Intermediate]** • **Target: The p-test for improper integrals**

Consider the infinite sequence defined by $a_n = \frac{3n^2 + 2n - 1}{n^2 + 5}$. To determine if this sequence converges, we examine its limit as $n \to \infty$. Which of the following statements is true about the limit of a_n?

a. The limit is 0.

b. The limit is 2.

c. The limit is 3.

d. The sequence diverges.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Combining partial fractions with improper integrals** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

The polar curve is given by the equation $r = 4\cos(\theta)$. What is the equivalent Cartesian equation for this curve?

a. $x^2 + y^2 = 4x$

b. $x^2 + y^2 = 16$

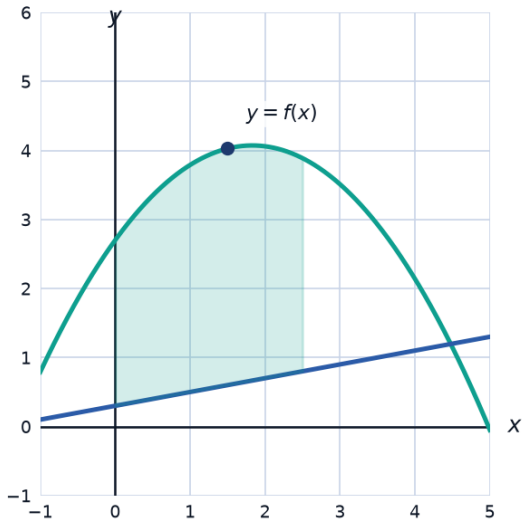
c. $y = 4x$

d. $x = 4\cos(y)$

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39 [Challenging] • Target: Three distinct linear factors • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Consider the infinite series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. For what value(s) of p does this series converge?



a. $p \leq 1$

b. $p < 1$

c. $p > 1$

d. $p = 1$

40 [Challenging] • Target: Visualizing convergence: area under the curve • Cross-Lesson: 1.01 Integration by Parts

Geometric series $\sum ar^n$ converges when:

a. $r > 1$

b. $|r| < 1$

c. $r < 0$

d. $r = 1$

41 [Challenging] • Target: Improper integrals with both an infinite bound and a discontinuity • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Remainder estimate for alternating series: $|R_n| \leq$

a. a_n

b. a_{n+1}

c. $\frac{1}{n}$

d. na_n

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42 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 3.01 Parametric Equations

What curve is described by the polar equation $r = 3$?

a. a line through the origin

b. a circle of radius 3 centred at the origin

c. a horizontal line $y = 3$

d. a vertical line $x = 3$

43 [Challenging] • Target: Partial fraction decomposition: distinct linear factors • Cross-Lesson: 3.02 Vector-Valued Functions

Find the area enclosed by the polar curve $r = 14$.

a. 196π

b. 14π

c. 28π

d. 98π

44 [Challenging] • Target: Partial fractions with a repeated linear factor • Cross-Lesson: 4.01 Polar Curves and Derivatives

Interval of convergence of $\sum \frac{x^n}{n}$ is:

a. $(-1, 1)$

b. $[-1, 1)$

c. $(-1, 1]$

d. $[-1, 1]$

45 [Challenging] • Target: Partial fractions with an irreducible quadratic factor • Cross-Lesson: 4.02 Area in Polar Coordinates

Classify $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$: absolutely convergent, conditionally convergent, or divergent?

a. Converges absolutely

b. Diverges

c. Converges conditionally

d. Option 0

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